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WEATHERGPT

Problem Statement ID – SIH26068
Problem Statement Title – Weather GPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information
Theme – Disaster Management
PS Category – Software
Team Name – CIPHER X

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WEATHERGPT

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The idea: turn official weather data into clear answers and useful next steps.

01 Hard-to-understand forecasts

Government forecasts often use technical terms that are difficult for everyday users to understand.

02 Unclear local warnings

Critical weather alerts may not reach people in a clear, simple and regional-language format.

03 Need for actionable advice

People need to know what the forecast means for them and what they should do next—not just rainfall percentages.

Our difference: explain the forecast clearly, connect it to the user's situation, and keep the information grounded in trusted weather data.

SYSTEM ARCHITECTURE

Weather APIs
NOAA / OpenWeatherMap

→

Data Ingestion
Collect + normalize

→

Weather Translator
Simple language

→

Verified LLM
+ guardrails

→

Alert / Cache
Warnings + SMS-ready

→

Web / Mobile UI
Chat, forecasts, alerts

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TECHNICAL APPROACH

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- **How the website works: ask → fetch trusted data → simplify → verify → alert → display**

Weather APIs
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Web / Mobile UI
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APP / WEBSITE FLOW

- User enters a question or selects a location.
- System retrieves forecast, warning, radar and climate data.
- Different weather formats are normalized into one structure.
- The translator converts technical data into simple language.

RELIABILITY + DELIVERY

- Verified AI answers only from retrieved weather data and official warnings.
- Guardrails keep the response grounded in the source data.
- Critical alerts are cached for weak connectivity.
- Web/mobile UI delivers forecasts, alerts and climate information.

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FEASIBILITY AND VIABILITY

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TECHNICAL FEASIBILITY

• Uses existing weather APIs such as NOAA and OpenWeatherMap.

• LLM explains forecast data instead of replacing the forecast model.

• Modular architecture allows APIs, AI, alerts and UI to be updated independently.

• Smart caching keeps essential information available when connectivity is weak.

VIABILITY

• Targets weather-sensitive users such as farmers, event planners and logistics fleets.

• The same core platform can scale to institutions and disaster-response use cases.

• API-based weather data + lightweight AI keeps the solution practical to operate.

• User feedback and verified data can continuously improve explanations and alerts.

HOW WE KEEP THE SYSTEM RELIABLE

• Ground every AI response in retrieved weather data and official warnings.

• Use official warning text as the fallback for critical information.

• Cache important forecasts and warnings so essential information remains accessible.

• Keep the system modular so individual components can be improved without rebuilding the whole platform.

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IMPACT AND BENEFITS

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IMPACT — WHAT CHANGES

- Faster access to understandable weather information.
- Better preparedness before severe weather events.
- Wider access for people who do not understand technical forecasts.
- Stronger disaster-management communication through timely alerts.

BENEFITS — WHAT USERS GAIN

- Time saved by getting instant answers instead of checking multiple sources.
- Lower weather-related losses for crops, logistics and events.
- Complex meteorological information made easier for non-experts.
- Forecasts, alerts and climate information brought together in one interface.

- **Goal: clearer information → earlier decisions → better preparedness**

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RESEARCH AND REFERENCES

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• **References used for the conversational weather and AI approach:**

1. **Hierarchical AI-Meteorologist: LLM-Agent System for Weather Forecast Reporting — arXiv**

   https://arxiv.org/html/2511.23387

2. **A Modular LLM-Agent System for Transparent Multi-Parameter Weather Interpretation — arXiv**

   https://arxiv.org/html/2512.11819

3. **Develop a Smart Weather Chatbot Using AutoGen**

   https://bryananthonio.com/

4. **Weather Chatbot — Multi-LLM Assistant (live demo)**

   https://weather-chatbot-llm.vercel.app/